

COMPUTER SCIENCE TRIPOS Part IB – 2024 – Paper 7

6 Formal Models of Language (pjb48)

Two friends, Andrew and Brian, live on the top of adjacent hills. In order to reduce their telephone bills, they decide to try communicating using a flag. They use one flag that can be held in one of four positions (Left, Right, Up, Down). To evaluate their proposed system they do a trial. Every 5 seconds Andrew holds the flag in a different position and Brian notes down where he thinks the flag is. The results of their trial are below:

		ANDREW'S FLAG POSITION			
		LEFT	RIGHT	UP	DOWN
BRIAN'S OBSERVATION	LEFT	2	0	1	1
	RIGHT	2	0	1	1
	UP	2	8	4	2
	DOWN	2	0	2	4

Entropy

- (a) Let X be a random variable for the flag positions produced by Andrew. Assuming the trial is probabilistically representative of the underlying distribution, write down the distribution of X over the discrete set of flag events $\mathcal{X} = \{\text{LEFT}, \text{RIGHT}, \text{UP}, \text{DOWN}\}$ [1 mark]

Answer: $\forall x \in X. p(x) = \frac{1}{4}$ [1 mark]

Entropy

- (b) What is the entropy, $H(X)$, of X in bits? Provide relevant equations **AND** describe in words what $H(X)$ represents in this scenario. [2 marks]

Answer: From here onwards small slips in arithmetic are not penalised if the equations are correct. $H(X) = \sum_{x \in \mathcal{X}} p(x) \log_2 \left(\frac{1}{p(x)} \right) = - \sum_{x \in \mathcal{X}} p(x) \log_2 p(x)$

In this case $\forall x \in X. p(x) = \frac{1}{4}$ so $H(X) = -\log_2 p(x) = \log_2 4 = 2$ [1 mark]

Something like: the average amount of uncertainty of which flag Andrew will produce [1 mark]

Joint entropy

- (c) Let Y be a random variable for the flag positions seen by Brian. Calculate the joint entropy $H(X, Y)$ in bits **AND** comment on whether this is the ideal joint entropy for this scenario. Provide relevant equations. [2 marks]

Answer: $H(X, Y) = - \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p(x, y) \log_2 p(x, y)$ $H(X, Y) = \frac{4*5}{32} + \frac{6*4}{16} + \frac{2*3}{8} + \frac{1*2}{4} = \frac{5}{8} + \frac{3}{2} + \frac{3}{4} + \frac{1}{2} = \frac{27}{8}$ [1 mark] No, ideal $H(X, Y) = 2$ when the information needed to specify both variables is the same as specifying just $H(X)$ [1 mark]

Conditional entropy

- (d) What is the conditional entropy $H(Y|X)$ in bits? Provide relevant equations **AND** describe in words what $H(Y|X)$ represents in this scenario. [2 marks]

Answer: $H(Y|X) = \sum_{x \in \mathcal{X}} p(x) H(Y|X = x) = - \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p(x, y) \log_2 p(y|x)$

$H(Y|X) = (\frac{1}{4})H(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}) + (\frac{1}{4})H(0, 0, 1, 0) + (\frac{1}{4})H(\frac{1}{8}, \frac{1}{8}, \frac{1}{2}, \frac{1}{4}) + (\frac{1}{4})H(\frac{1}{8}, \frac{1}{8}, \frac{1}{2}, \frac{1}{4})$

— *Solution notes* —

$$H(Y|X) = \frac{11}{8} \text{ [1 mark]}$$

Something like: is the average amount of information needed to know what Brian wrote down given the flag that Andrew transmitted [1 mark]

Mutual
Information

- (e) Calculate the mutual information between X and Y in bits **AND** explain what this tells us about their communication system. Provide relevant equations. [3 marks]

Answer: $I(X;Y) = H(Y) - H(Y|X) = \frac{7}{4} - \frac{11}{8} = \frac{3}{8}$ [1 mark] It is a measure of the reduction in uncertainty of Andrew's flag due to what Brian wrote down [1 mark] and vice versa [1 mark] that is the amount of information one random variable contains about another.

Channel Capacity

- (f) Provide an attested lower bound on the channel capacity for Andrew and Brian's communication system. Explain your answer providing any relevant equations. [3 marks]

Answer: The capacity of a channel is the maximum of the mutual information of X and Y over all input distributions of the input $p(X)$ [1 mark]
 $C = \max_{p(X)} I(X;Y)$ [1 mark]
An attested lower bound, then, is any calculated mutual information, including the one from the question above. [1 mark]

Interpreting
metrics

- (g) Do you think this will be an effective way for Andrew and Brian to communicate? Explain your answer. [2 marks]

Answer: Something like: depends on the intended distribution of X [1 mark] . For this particular distribution mutual information is low and conditional entropy $H(Y|X)$ is high and the message would be not successfully transmitted [1 mark]

Interpreting
metrics

- (h) Andrew suggests using two positions for the flags instead of four. What do you think about this suggestion? What two positions would you choose and why? Explain how you could quantify the benefit (a numerical answer is not required). [5 marks]

Answer: Requires valid interpretation of the table and the metrics. Explanation that we want to increase the mutual information [1 mark] Comments might include: not using the left flag position [1 mark] merging up and right flag position [1 mark] explanation of how you might rewrite the table and recalculate $I(X;Y)$ [2 marks]
